MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 01 (SURFACE AREAS & VOLUMES)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show clear steps in calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. A hollow spherical shell of internal and external diameters 4 cm and 8 cm is melted into a cone of base diameter 8 cm. The height of the cone is: **[1]**

(a) (a) 12 cm (b) 14 cm (c) 15 cm (d) 18 cm

2. A solid piece of iron of dimensions 49 cm × 33 cm × 24 cm is moulded into a solid sphere. The radius of the sphere is: **[1]**

(a) (a) 21 cm (b) 28 cm (c) 35 cm (d) 14 cm

3. The ratio of lateral surface area to the total surface area of a cylinder with base diameter 1.6 m and height 20 cm is: **[1]**

(a) (a) 1 : 5 (b) 1 : 7 (c) 1 : 8 (d) 1 : 9

4. A solid consists of a circular cylinder surmounted by a right circular cone of height h. If the total volume of the solid is 3 times the volume of the cone, the height of the cylinder is: **[1]**

(a) (a) h (b) $\frac{2h}{3}$ (c) $\frac{3h}{2}$ (d) $2h$

5. **Assertion (A):** Two solid metallic right circular cones of equal height and base radii 3 cm and 4 cm are melted and recast into a solid sphere of radius 5 cm. The cones are of height 20 cm each.
Reason (R): If two right circular cones of height h and base radii r_1, r_2 are recast into a sphere of radius R, then $h = \frac{4R^3}{r_1^2 + r_2^2}$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. The radii of the bases of a cylinder and a cone are in the ratio 3 : 4 and their heights are in the ratio 2 : 3. What is the ratio of their volumes? **[2]**

7. If the heights of two right circular cones are in the ratio 1 : 2 and the perimeters of their bases are in the ratio 3 : 4, what is the ratio of their volumes? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. Two identical solid cubes of side a are joined end to end. Find the total surface area of the resulting cuboid. **[3]**

9. An empty cone is of radius 3 cm and height 12 cm. Ice-cream is filled in it so that lower part which is $\frac{1}{6}$ of the volume is unfilled, but a hemispherical scoop of radius 3 cm is on top. Find the volume of ice-cream. (Take $\pi = 3.14$) **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (5 Marks) [5 Marks]

10. A cistern, internally measuring 150 cm $\times$ 120 cm $\times$ 110 cm, has 129, 600 cm^3 of water in it. Porous bricks of dimensions 22.5 cm $\times$ 7.5 cm $\times$ 6.5 cm are placed in the water. Each brick absorbs $\frac{1}{17}$th of its own volume of water. Find how many bricks can be put in the cistern without water overflowing to the brim. **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

11. Case Study: The Commonwealth Games Victory Podium

Organisers constructed a 3-tier wooden victory podium for medal ceremonies consisting of three joined rectangular cuboids placed side-by-side on the ground: 1st Position Block (Gold, Center) of size 40 cm $\times$ 50 cm $\times$ 40 cm, 2nd Position Block (Silver, Left) of size 40 cm $\times$ 50 cm $\times$ 24 cm, and 3rd Position Block (Bronze, Right) of size 40 cm $\times$ 50 cm $\times$ 12 cm.

(i) Find the surface area of the front vertical face of the entire victory stand.
(ii) Find the volume of the block designated for the 2nd position athlete (Silver).
(iii) Find the volume of the block designated for the 1st position winner (Gold).
(iv) Find the volume of the block designated for the 3rd position athlete (Bronze).

[4]

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— END OF QUESTION PAPER —

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